

A Foundation Model for Cross-Band CSI Reconstruction

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Abstract—Acquiring dense high-frequency channel state information (CSI) is costly in multi-band low-altitude wireless systems because pilot resources are limited and channel dimensions change with the carrier frequency, bandwidth, and antenna array size. We address cross-band CSI reconstruction, which recovers dense target-band CSI from dense source-band CSI and sparse, noisy target-band pilots. We propose a foundation model that represents every band in a common power–angle–delay spectrum and uses radio-frequency metadata as auxiliary conditioning for an encoder–decoder. The model uses pilot-guided cross-attention to fuse source-band structure with target-band pilot, allowing one model to handle heterogeneous band pairs. It is trained by pilot-densification pretraining followed by supervised cross-band fine-tuning. On ray-tracing, 3GPP, and DeepMIMO datasets, the model lowers average normalized mean-square error by 6.1 dB over the state-of-the-art pair-specific baseline. It also transfers to two unseen pairs without paired fine-tuning, achieving 7.5 and 7.1 dB gains over fully supervised baselines, and remains effective when target pilots are noisy.

Index Terms—mmWave, channel estimation, channel extrapolation, Transformer

I. INTRODUCTION

Future low-altitude wireless networks are expected to operate in multiple frequency bands, including sub-6 GHz and millimeter-wave (mmWave) frequency bands. This trend makes accurate channel state information (CSI) essential, but also renders dense high-frequency CSI expensive to acquire, especially for high frequency bands: the larger spatial-frequency dimensions inflate pilot and feedback overhead, and the higher-dimensional channels are sounded by proportionally fewer pilots under a fixed pilot budget, raising sensitivity to pilot sparsity and noise [1]. A natural way to reduce this cost is to reuse the channel of a lower-frequency, more easily sounded source band. Out-of-band information has long been used this way: to steer mmWave beam selection [2] or to extrapolate the channel across frequency and avoid explicit feedback [3], and learning-based methods more recently extend the idea to mapping channels across antenna and frequency configurations [4]. The shared premise is that bands sharing a propagation environment retain the same dominant delay and angular structure, so a lower band carries a useful prior for reconstructing the target-band CSI.

However, exploiting such information is difficult because the shared structure is only partial: array responses, delay resolutions, path gains, and pilot observations all vary with carrier frequency, bandwidth, and antenna configuration [2], [5]. A

source-band CSI matrix therefore cannot be mapped directly to the target band, and sparse target-band pilots are still needed to anchor the reconstruction. Classical interpolation uses only pilots and ignores the cross-band prior, while most learning-based CSI work targets feedback compression [6], which does not address reconstruction from sparse cross-band observations and faces persistent generalization and configuration-dependence challenges [7].

These requirements motivate a foundation-model approach: a single model should learn reusable channel representations and remain aware of varying radio-frequency (RF) configurations. Recent wireless foundation models [8] pursue this direction through large-scale self-supervised pretraining: For instance, LWM learns task-agnostic channel embeddings [9], WiFo targets one-for-all channel prediction across heterogeneous configurations [10], and ContraWiMAE learns masked-contrastive channel representations [11]. These works show the promise of pretrained, zero-shot channel representations, but mainly target general representation or prediction. Cross-band reconstruction is a related yet less explored setting, where a dense source-band channel and a few noisy target-band pilots are jointly available for target-band CSI recovery.

In this work, we propose an encoder–decoder for cross-band CSI reconstruction with source-band CSI and target-band pilots as inputs. The proposed method combines a common delay–angle representation, pilot-guided source–target fusion, and RF metadata conditioning in one network. Our main contributions are as follows.

- **A foundation model for heterogeneous cross-band reconstruction.** We place heterogeneous CSI on a common delay–angle grid and fuse source CSI with sparse target pilots in one backbone, while RF prompts provide auxiliary band descriptors for heterogeneous configurations.
- **A task-aligned two-stage training scheme.** A pilot-densification pretext is followed by cross-band fine-tuning, first learning to densify sparse self-pilots and then adapting the same architecture to source–target fusion.
- **Extensive experimental validation.** Across six band pairs, the proposed method lowers normalized mean-square error (NMSE) by 6.1 dB on average over the state-of-the-art full-shot baseline, transfers to unseen band pairs zero-shot, and remains robust under noisy pilots.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Channel Model

We consider a base station (BS) operating over a set of carrier bands $\mathcal{B}$. For each band $b \in \mathcal{B}$, the BS is equipped with a band-specific uniform planar array (UPA) and orthogonal frequency division multiplexing (OFDM) numerology. These UPAs share the same BS site but may have different array dimensions and carrier-dependent element spacings. Specifically, the UPA for band b has $N_y^{(b)} \times N_z^{(b)}$ elements, with $N_t^{(b)} = N_y^{(b)} N_z^{(b)}$, and its OFDM grid has $N_c^{(b)}$ subcarriers with spacing $\Delta f^{(b)}$. The baseband frequency of subcarrier n is $f_n^{(b)} = (n - \lfloor N_c^{(b)}/2 \rfloor) \Delta f^{(b)} + f_c^{(b)}$, the center wavelength is $\lambda_c^{(b)} = c_0/f_c^{(b)}$ with c_0 the speed of light, and the UPA takes half-wavelength spacing $d_y^{(b)} = d_z^{(b)} = \lambda_c^{(b)}/2$. A path ℓ has delay τ_ℓ and direction cosines $u_\ell = \sin \theta_\ell \cos \phi_\ell$, $v_\ell = \sin \theta_\ell \sin \phi_\ell$, collected as $\mathbf{o}_\ell = (u_\ell, v_\ell)$. The subcarrier channel and its steering vector are

$$\mathbf{h}_n^{(b)} = \sum_{\ell=1}^L \alpha_\ell^{(b)} e^{-j2\pi f_n^{(b)} \tau_\ell} \mathbf{a}_n^{(b)}(\mathbf{o}_\ell), \quad (1)$$

$$[\mathbf{a}_n^{(b)}(\mathbf{o}_\ell)]_{m_y, m_z} = \frac{1}{\sqrt{N_t^{(b)}}} \exp\left(-j2\pi \frac{(m_y d_y^{(b)} u_\ell + m_z d_z^{(b)} v_\ell) f_n^{(b)}}{c_0}\right), \quad (2)$$

with $m_y = 0, \dots, N_y^{(b)} - 1$ and $m_z = 0, \dots, N_z^{(b)} - 1$. The UPA response is vectorized to form $\mathbf{h}_n^{(b)} \in \mathbb{C}^{N_t^{(b)}}$, and stacking all subcarriers gives $\mathbf{H}^{(b)} \in \mathbb{C}^{N_c^{(b)} \times N_t^{(b)}}$. Co-located bands share almost similar path geometry $\{(\tau_\ell, \mathbf{o}_\ell)\}$, while the path gains $\alpha_\ell^{(b)}$, array dimensions, and RF parameters are band-dependent.

The target band is observed only through sparse, noisy pilots. Under a uniform lattice pattern with strides (κ_f, κ_a) ,

$$\bar{\mathbf{H}}_{pilot} = \mathbf{M}_{\kappa_f, \kappa_a} \odot (\bar{\mathbf{H}}_t + \mathbf{N}), \quad (3)$$

where $\bar{\mathbf{H}}_t$ is the normalized target CSI, $\mathbf{N}$ is pilot noise at a prescribed signal-to-noise ratio (SNR), and the binary mask $\mathbf{M}_{\kappa_f, \kappa_a}$ retains every κ_f -th subcarrier and κ_a -th antenna (entry 1 when $n \bmod \kappa_f = 0$ and $m \bmod \kappa_a = 0$, else 0), giving pilot density $1/(\kappa_f \kappa_a)$. Each band has RF configuration $(f_c^{(b)}, N_c^{(b)}, \Delta f^{(b)}, N_t^{(b)})$, with network prompt $\mathbf{r}^{(b)}$ specified in Section III.

B. Problem Formulation

Co-located bands that share a propagation environment retain the same geometry-induced delay-angle structure while differing in carrier frequency, array size, subcarrier grid, and per-path gains, as mentioned in Section I. We exploit this property in the following setting: an easily sounded *source* band s assists the acquisition of a costly *target* band t , and our goal is to recover the dense target CSI from the source observation together with only a few target-band pilots.

Formally, given the normalized source CSI $\bar{\mathbf{H}}_s$, the sparse noisy target pilots $\bar{\mathbf{H}}_{pilot}$, and the RF prompts $\mathbf{r}_s, \mathbf{r}_t$, we

estimate the dense target CSI $\bar{\mathbf{H}}_t$. We learn a mapping $\mathcal{F}_\Theta$ that is shared across all pairs,

$$\hat{\bar{\mathbf{H}}}_t = \mathcal{F}_\Theta(\bar{\mathbf{H}}_s, \bar{\mathbf{H}}_{pilot}, \mathbf{r}_s, \mathbf{r}_t), \quad (4)$$

and optimize its parameters Θ by minimizing the expected target CSI reconstruction error, with the training objective detailed in Section III.

III. PROPOSED METHOD

In this section, we present the proposed encoder-decoder network for cross-band CSI reconstruction. The model architecture, including the overall data flow and the encoder/decoder block designs, is shown in Fig. 1.

A. Unified Power-Angle-Delay Spectrum Representation

To place the band-dependent CSI matrices $\mathbf{H}^{(b)}$ on a common grid, we map each CSI matrix to a fixed-size power-angle-delay spectrum (PADS). For each band, we build a delay dictionary $\mathbf{W}_d^{(b)} \in \mathbb{C}^{H \times N_c^{(b)}}$ and an angular dictionary $\mathbf{W}_a^{(b)} \in \mathbb{C}^{N_t^{(b)} \times W}$. Let $T_{\max}^{(b)} = 1/\Delta f^{(b)}$, $\tau_i = iT_{\max}^{(b)}/H$, $u_j = 2j/W - 1$, and $q_m = m - (N_t^{(b)} - 1)/2$. The dictionaries are constructed as

$$[\mathbf{W}_d^{(b)}]_{i,n} = H^{-1/2} \exp(j2\pi \tau_i f_n^{(b)}), \quad (5)$$

$$[\mathbf{W}_a^{(b)}]_{m,j} = (N_t^{(b)})^{-1/2} \exp(-j\pi q_m u_j), \quad (6)$$

Given the dataset-level CSI and PADS scales $s_H^{(b)}$ and $s_P^{(b)}$, the model input is

$$\mathbf{P}^{(b)} = \mathbf{W}_d^{(b)} (\mathbf{H}^{(b)} / s_H^{(b)}) \mathbf{W}_a^{(b)}, \quad (7)$$

$$\mathbf{X}^{(b)} = [\Re(\mathbf{P}^{(b)} / s_P^{(b)}), \Im(\mathbf{P}^{(b)} / s_P^{(b)})] \in \mathbb{R}^{2 \times H \times W}. \quad (8)$$

The scale statistics are computed from the datasets as described in Section IV-A. This gives all bands a shared delay-angle input grid while preserving propagation sparsity.

B. Band-Aware Encoder

We first tokenize the input. A convolutional stem down-samples the PADS $\mathbf{X} \in \mathbb{R}^{2 \times H \times W}$ by $4\times$, and a patch embedding with patch size $p_h \times p_w$ projects the result into $N = HW/p_h p_w$ tokens of dimension D . We prepend a learnable class token $\mathbf{t}_{cls} \in \mathbb{R}^D$ and add a fixed two-dimensional sinusoidal positional encoding $\mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D}$, yielding the token sequence $\mathbf{Z}^0 \in \mathbb{R}^{(N+1) \times D}$,

$$\mathbf{Z}^0 = [\mathbf{t}_{cls}; \text{PatchEmbed}(\text{Stem}(\mathbf{X}))] + \mathbf{E}_{pos}, \quad (9)$$

where, for a patch at grid position (p, q) , the positional encoding concatenates one-dimensional sinusoidal codes of its row and column, $[\mathbf{E}_{pos}]_{p,q} = [\mathbf{e}(p); \mathbf{e}(q)]$, with $\mathbf{e}(x) \in \mathbb{R}^{D/2}$ defined by $e_{2i}(x) = \sin(x\omega_i)$, $e_{2i+1}(x) = \cos(x\omega_i)$ and $\omega_i = 10000^{-4i/D}$.

To make these tokens band-aware, we summarize the RF configuration of band b as the prompt $\mathbf{r}^{(b)} \in \mathbb{R}^4$,

$$\mathbf{r}^{(b)} = [\log f_c^{(b)}, \log N_c^{(b)}, \log \Delta f^{(b)}, \log N_t^{(b)}], \quad (10)$$

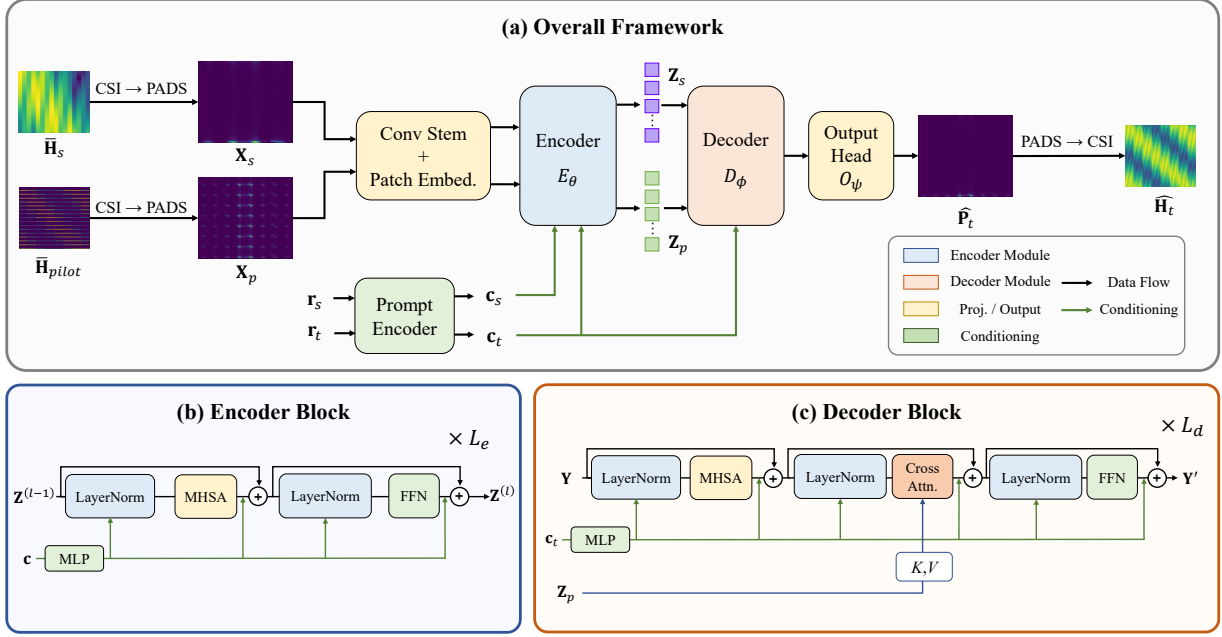


Fig. 1. Model architecture of the proposed method, including the overall framework and the encoder/decoder block designs.

and lift it by random Fourier features and a two-layer multi-layer perceptron (MLP) into a conditioning vector $\mathbf{c}^{(b)} \in \mathbb{R}^D$. Rather than appending $\mathbf{c}^{(b)}$ as an extra token, every Transformer block injects it through adaptive layer normalization (AdaLN) [12],

$$\text{AdaLN}(\mathbf{Y}; \mathbf{c}) = (1 + \mathbf{g}(\mathbf{c})) \odot \text{LayerNorm}(\mathbf{Y}) + \mathbf{b}(\mathbf{c}), \quad (11)$$

where the scale $\mathbf{g}(\mathbf{c}) \in \mathbb{R}^D$ and shift $\mathbf{b}(\mathbf{c}) \in \mathbb{R}^D$ are linear functions of the RF condition. The carrier frequency, bandwidth, and array size therefore reshape the feature statistics for each sub-layer.

Each of the L_e encoder blocks adopts the pre-norm form, applying AdaLN-modulated multi-head self-attention (MHSA) [13] and then a SwiGLU feed-forward network (FFN), each wrapped in a gated residual connection. For block ℓ with input $\mathbf{Z}^{\ell-1}$,

$$\mathbf{Z}' = \mathbf{Z}^{\ell-1} + \mathbf{s}_a \odot \text{MHSA}(\text{AdaLN}(\mathbf{Z}^{\ell-1}; \mathbf{c})), \quad (12)$$

$$\mathbf{Z}^\ell = \mathbf{Z}' + \mathbf{s}_f \odot \text{FFN}(\text{AdaLN}(\mathbf{Z}'; \mathbf{c})), \quad (13)$$

where the residual gates $\mathbf{s}_a, \mathbf{s}_f \in \mathbb{R}^D$, together with the scale and shift of the two AdaLN instances, are all produced from $\mathbf{c}$ by a linear layer. The encoder output is the final-block sequence $\mathbf{Z} = \mathbf{Z}^{L_e}$.

The encoder E_θ handles both inputs: it processes the full source PADS $\mathbf{X}_s$ under the source condition and the sparse target-pilot PADS $\mathbf{X}_p$ under the target condition,

$$\mathbf{Z}_s = E_\theta(\mathbf{X}_s, \mathbf{c}_s), \quad \mathbf{Z}_p = E_\theta(\mathbf{X}_p, \mathbf{c}_t), \quad (14)$$

with $\mathbf{Z}_s, \mathbf{Z}_p \in \mathbb{R}^{(N+1) \times D}$. Weight sharing places the source and pilot features in a common token space that the decoder can fuse directly, while the prompt specializes the same weights to each band.

C. Pilot-Guided Decoder

The decoder starts from the encoded source sequence $\mathbf{Y}^0 = \mathbf{Z}_s$ and uses the target-pilot sequence $\mathbf{Z}_p$ as a cross-attention context. Since the output belongs to the target band, every decoder block is conditioned on $\mathbf{c}_t$.

Each of the L_d decoder blocks has three sub-layers, all conditioned on $\mathbf{c}_t$: MHSA, cross-attention, and a SwiGLU FFN. The MHSA and FFN take the same form as in the encoder. The cross-attention module draws queries from the current source stream $\mathbf{Y}$ and keys/values from the pilots $\mathbf{Z}_p$,

$$\mathbf{Y} \leftarrow \mathbf{Y} + \mathbf{s}_x \odot \text{CrossAttn}(\text{AdaLN}(\mathbf{Y}; \mathbf{c}_t), \mathbf{Z}_p), \quad (15)$$

where, for a single head,

$$\text{CrossAttn}(\mathbf{Y}, \mathbf{Z}_p) = \text{Softmax}(\mathbf{Y}\mathbf{W}_Q \mathbf{W}_K^\top \mathbf{Z}_p^\top / \sqrt{d}) \mathbf{Z}_p \mathbf{W}_V \quad (16)$$

The source stream thus queries target-band evidence while the pilot representation stays fixed. After the final block, an output head O_ψ reshapes the N patch tokens $\mathbf{Y}^{L_d}_{1:N}$ and upsamples it to the dense target PADS $\hat{\mathbf{P}}_t \in \mathbb{R}^{2 \times H \times W}$, which the fixed pseudo-inverse dictionaries map back to CSI:

$$\hat{\mathbf{H}}_t = (\mathbf{W}_d^{(t)})^\dagger \hat{\mathbf{P}}_t (\mathbf{W}_a^{(t)})^\dagger. \quad (17)$$

D. Two-Stage Training

Both stages draw pilots from the uniform lattice of Section II and they reuse the same shared encoder E_θ , decoder D_ϕ , and output head O_ψ , differing only in how these modules are fed. All parameters are updated in both stages.

The first stage is *pilot-densification pretraining*. Given a dense PADS sample $\mathbf{X}^{(b)}$, we sample a pilot lattice from its CSI at a randomly chosen density, add Gaussian noise, and

transform the sparse observation back to PADS as $\mathbf{X}_{pilot}^{(b)}$. The decoder receives the same tokens as both query and context,

$$\hat{\mathbf{X}}^{(b)} = O_\psi(D_\phi(\mathbf{Z}^{(b)}, \mathbf{Z}^{(b)}, \mathbf{c}^{(b)})), \mathbf{Z}^{(b)} = E_\theta(\mathbf{X}_{pilot}^{(b)}, \mathbf{c}^{(b)}). \quad (18)$$

The target is the dense PADS $\mathbf{X}^{(b)}$, and the loss is its PADS-domain normalized mean square error (NMSE),

$$\mathcal{L}_{pt} = \sum_i \|\hat{\mathbf{X}}_i - \mathbf{X}_i\|_F^2 / \sum_i \|\mathbf{X}_i\|_F^2. \quad (19)$$

The second stage is *cross-band fine-tuning*. For each paired sample, the source band provides the dense PADS $\mathbf{X}_s$, while the target band provides the sparse target-pilot PADS $\mathbf{X}_p$ and the dense CSI supervision,

$$\hat{\mathbf{X}}_t = O_\psi(D_\phi(E_\theta(\mathbf{X}_s, \mathbf{c}_s), E_\theta(\mathbf{X}_p, \mathbf{c}_t), \mathbf{c}_t)), \quad (20)$$

Then $\hat{\mathbf{X}}_t$ is mapped to CSI by the inverse dictionaries and the loss $\mathcal{L}_{ft}$ is the same NMSE evaluated in the CSI domain, between $\hat{\mathbf{H}}_t$ and $\bar{\mathbf{H}}_t$. For multi-pair training, each batch contains a single frequency pair.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

Datasets and splits. We evaluate the proposed method on CSI datasets from ray-tracing (RT)¹, 3GPP stochastic by QuaDRiGa [14], and DeepMIMO channels [15], with scene lists, sample counts, and CSI configurations summarized in Table I. In the RT subset, Biomedical Engineering Buildings (BME), Dong 4 Area (D4), and East Middle Buildings (EMB) are derived from Shanghai Jiao Tong University scenes whose maps are extracted from OpenStreetMap [16], Los Angeles (LA) is self-built, and all RT CSI samples are generated by Sionna RT [17]. All the CSI samples correspond to far-field channels [18]. The two stages use separate pools: pretraining uses single-band CSI from 27 scenes (EMB and DeepMIMO city 0 held out for validation); fine-tuning uses six frequency pairs from 22 scenes, and testing uses five held-out scenes. Per-scene, per-band root mean square (RMS) scales for complex CSI and for the real/imaginary PADS are computed once and reused across training, validation, and testing.

Task and evaluation. Each test sample contains dense source-band CSI $\mathbf{H}_s$ and noisy sparse target-band pilots $\bar{\mathbf{H}}_{pilot}$; the output is dense target-band CSI $\hat{\mathbf{H}}_t$. Fine-tuning uses six supervised frequency pairs: 2.4→15, 2.4→28, 3.5→15, 3.5→28, 15→28, and 15→40 GHz. Pair-specific zero-shot settings are given per experiment. Unless otherwise noted, target pilots use a uniform lattice with frequency stride 8 and antenna stride 1 (12.5% density), and pilot noise is 10–20 dB SNR in training. We report CSI-domain NMSE at 15 dB test SNR, with all methods on the same test subset and results averaged over each pair’s test scenes.

Baselines. All baselines output the dense target CSI $\hat{\mathbf{H}}_t$. **Linear interpolation** and **cubic spline interpolation** are non-learned pilot-only methods: their inputs are the noisy sparse

target pilots $\bar{\mathbf{H}}_{pilot}$ and the pilot mask, and they independently interpolate missing subcarriers for each antenna. **U-Net** [19] is a learned pilot-only network with a three-level encoder–decoder and is trained to predict the target CSI. **CsiNet** [6] is adapted to cross-band reconstruction by using the source CSI $\mathbf{H}_s$ and target pilots as two input branches: a CsiNet-style source encoder and a target-pilot encoder produce latent vectors, which are then fused and decoded. **LWM** [9] uses source CSI and target pilots converted to patch tokens, and the LWM backbone is fine-tuned together with a cross-attention head to recover the target CSI. The proposed method uses the same source CSI and target pilots as the source-aware baselines, with RF prompts. Learned baselines use the same pilot mask and SNR, and target supervision. They are optimized with AdamW under CSI-domain NMSE, with checkpoints selected on the validation split. To give the baselines a favorable setting, we allow target-scene supervised training: for each cross-band test scene, samples are split 70/10/20 into train/validation/test. A separate baseline is trained on the 70% training split for each frequency pair and scene, optimized with AdamW under CSI-domain NMSE. All methods are evaluated on the same 20% test subset.

B. Implementation Details

All experiments use the network configuration in Table II, whose 40.7M parameters cover the tokenizer, RF prompt encoder, Transformer encoder, decoder, and output head. Table III lists the two-stage training hyperparameters, optimized with AdamW under cosine learning-rate decay, gradient clipping at 1.0, and fp32 precision. The fine-tuning ranges reflect per-pair selection on the validation split. Experiments run on a server with two AMD EPYC 9454 48-Core CPUs and four NVIDIA GeForce RTX 4090 GPUs.

C. Numerical Results

We first evaluate cross-band CSI reconstruction, and then examine pilot-noise robustness, zero-shot frequency-pair transfer, ablation study, and inference cost.

Table IV compares all methods on six supervised frequency pairs at a 15 dB pilot SNR. Our method achieves the best NMSE on every pair, improving over the most competitive baseline by 6.1 dB on average. The gains are smaller when the full-shot U-Net is already strong, but consistently show that source CSI provides complementary structure beyond target pilots alone. The results also highlight the need for both cross-band information and RF-conditioned fusion. Linear interpolation is competitive on easier 15 GHz targets but degrades as bandwidth and array size grow, and U-Net cannot exploit source-band CSI. CsiNet and LWM use source CSI, yet their inconsistent gains show that cross-band information must be fused with band awareness rather than treated as generic input.

Fig. 2 measures pilot-noise robustness by sweeping the target-pilot SNR from 0 to 20 dB on 2.4 → 15 and 2.4 → 28 GHz, while keeping the model parameters, pilot mask, and test subset fixed. The proposed method stays best across the

¹<https://github.com/Zetahp/RT-Datasets>

TABLE I
DATASET COMPOSITION, SAMPLES PER BAND, AND CSI CONFIGURATION.

Source	Dataset scenes	Band (GHz)	Samples	N_c	N_t	Δf (kHz)	UPA	Test scene
RT	BME, D4, EMB, LA	2.4	203K	64	16	60	4×4	EMB
		3.5	203K	64	16	90	4×4	
		15	203K	128	32	90	8×4	
		28	203K	128	64	180	8×8	
		40	203K	128	64	180	8×8	
3GPP	UMa, RMa, InH, InF, UMi	2.4	300K	64	16	60	4×4	InH
		3.5	300K	64	16	90	4×4	
		15	300K	128	32	90	8×4	
		28	300K	128	64	180	8×8	
DeepMIMO	city 0–19 except 4; ASU campus at 3.5 GHz	3.5	1,385K	64	16	90	4×4	city 0/5/15
		28	1,303K	128	64	180	8×8	

TABLE II
NETWORK CONFIGURATION.

Item	Value
PADS size $2 \times H \times W$	$2 \times 256 \times 256$
RF prompt dim.	4
Embed. dim. D	384
Enc. / Dec. depth L_e / L_d	6 / 6
Attn. heads	6
Stem downsampling	$4 \times$
Patch size $p_h \times p_w$	8×8
Params.	40.7M

TABLE III
TRAINING CONFIGURATION.

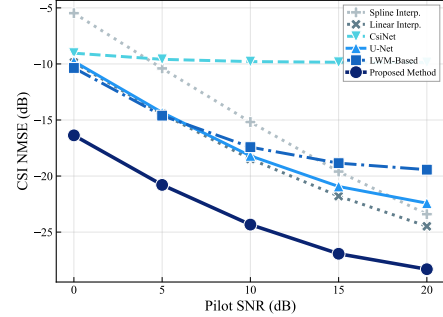
Item	Pretraining	Fine-tuning
Loss	PADS-domain NMSE	CSI-domain NMSE
Epochs	20	150
Batch size	512	256
Learning rate	2×10^{-4}	1.0×10^{-4}
Weight decay	0.05	0.01
Warmup steps	2000	500
Pilot stride (κ_f, κ_a)	$\{1, 4, 8, 16, 32\} \times 1$	8×1
Pilot SNR	10–20 dB	10–20 dB

TABLE IV
MAIN COMPARISON ACROSS FREQUENCY PAIRS.

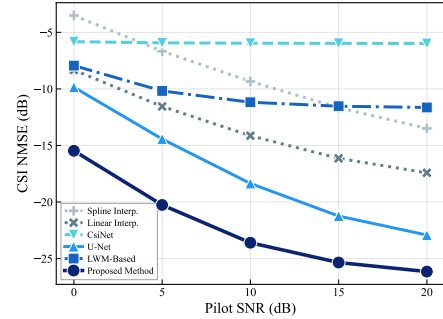
Method	2.4→15	2.4→28	3.5→15	3.5→28	15→28	15→40
Linear	-21.79	-16.14	-21.79	-15.71	-16.14	-9.38
Spline	-19.61	-11.66	-19.61	-10.31	-11.66	-4.26
CsiNet	-9.85	-5.97	-9.85	-6.89	-6.58	-1.06
U-Net	-20.93	-21.24	-20.77	-19.23	-21.29	-19.16
LWM	-18.85	-11.53	-17.61	-9.22	-11.31	-9.43
Proposed Method	-26.93	-25.34	-26.98	-28.98	-25.58	-27.03

range. At low SNRs all methods are limited by noise; at high SNRs the error is dominated by missing target samples, so interpolation and pilot-only networks approach a pilot-density ceiling, while the gap our method retains shows that cross-band geometry adds resolution beyond what sparse pilots provide, especially for the higher-dimensional 28 GHz target.

Fig. 3 evaluates whether the model can transfer to unseen source–target frequency pairs. Our method is fine-tuned only on the six supervised pairs in Table IV and is tested on $2.4 \rightarrow 40$ and $3.5 \rightarrow 40$ GHz, with no paired samples from these pairs used during fine-tuning. Although the baselines are trained full-shot on the evaluated pairs, the proposed method still improves by 7.5 and 7.1 dB over the strongest baseline, reaching -27.0 dB on both. Since 40 GHz is seen during



(a) $2.4 \rightarrow 15$ GHz



(b) $2.4 \rightarrow 28$ GHz

Fig. 2. Robustness of cross-band reconstruction to pilot noise.

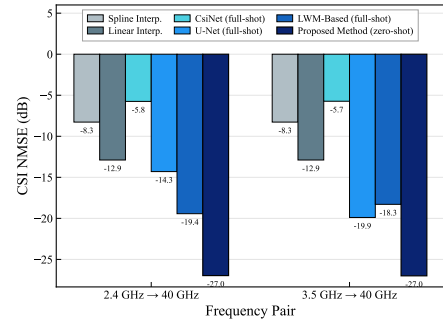


Fig. 3. Zero-shot transfer on unseen frequency pairs.

training only through $15 \rightarrow 40$, the model recombines a known target condition with known source conditions it never observed together rather than extrapolating to a new band.

For ablation, Fig. 4 removes one component at a time

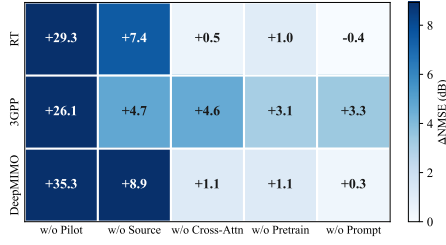


Fig. 4. Component ablation study.

TABLE V
INFERENCE COST.

Method	Params (M)	FLOPs (G)	Latency (ms)	Peak Mem. (MB)
CsiNet	18.6	0.10	0.6	213
U-Net	1.9	2.31	1.4	464
LWM	3.1	5.45	10.7	6883
Proposed Method	40.7	6.26	9.6	505

under the same split, pilot mask, and SNR, and reports the degradation across data sources. The two necessary inputs dominate everywhere: removing the target pilots costs 26–35 dB and removing the source CSI a further 4.7–8.9 dB. Removing source CSI reduces the model to a pilot-only estimator, whose accuracy matches the dedicated U-Net baseline in Table IV, confirming the drop reflects the genuine value of cross-band information. The source term helps most where the two bands share clean geometry (RT and DeepMIMO city, +7.4 and +8.9 dB) and least on the stochastic 3GPP InH channel. Cross-attention, RF prompting, and pretraining show the opposite result: each is near-neutral on RT and DeepMIMO but contributes 3.1–4.6 dB on 3GPP. Their small gains on RT and DeepMIMO do not indicate redundancy: on these channels pilots and source CSI already drive the model near a reconstruction ceiling, leaving little to remove, whereas on the hardest InH channel they recover 3.1–4.6 dB. The modules are most helpful when the channel is more complex.

Table V reports the inference cost on the $3.5 \rightarrow 28$ GHz task in fp32 on a single RTX 4090 at batch size 1, where peak memory is the maximum GPU memory allocated during one forward pass. The convolutional baselines have the lowest latency but a clear accuracy tradeoff in Table IV. Among the Transformer-scale models, the proposed method runs faster than LWM and uses less peak memory despite its larger parameter count, as the compact 8×8 token grid from the convolutional stem bounds attention cost and activation memory.

V. CONCLUSION

In this paper, we presented a foundation model for cross-band CSI reconstruction with source-band CSI and sparse target-band pilots as inputs. By aligning heterogeneous bands in a common PADS domain, fusing source and pilot features, and using RF metadata as auxiliary conditioning, the model learns one reconstruction function across multiple frequency pairs. Experiments show a 6.1 dB average improvement over

the strongest supervised baseline and strong zero-shot transfer to unseen source–target frequency pairs. These results suggest that common PADS-based representations with RF conditioning can reduce pair-specific adaptation while preserving practical inference latency and peak memory. Future work will relax the requirement of synchronized source-band CSI and validate the approach on measured multi-band channels.

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